

Math 526, F09
Quiz 1

Name: Answer
SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (10 points) Given $\mathbf{v} = (1, 1)$ and $\mathbf{w} = (\sqrt{3} + 1, \sqrt{3} - 1)$.
(a) Compute $\|\mathbf{v}\|$, $\|\mathbf{w}\|$, and $\mathbf{v} \cdot \mathbf{w}$.

$$\text{Solution: } \|\mathbf{v}\| = \sqrt{\mathbf{v} \cdot \mathbf{v}} = \sqrt{1^2 + 1^2} = \sqrt{2}$$

$$\|\mathbf{w}\| = \sqrt{\mathbf{w} \cdot \mathbf{w}} = \sqrt{(\sqrt{3}+1)^2 + (\sqrt{3}-1)^2}$$

$$= \sqrt{(3+2\sqrt{3}+1) + (3-2\sqrt{3}+1)}$$

$$= \sqrt{8} = 2\sqrt{2}$$

$$\mathbf{v} \cdot \mathbf{w} = (1, 1) \cdot (\sqrt{3}+1, \sqrt{3}-1)$$

$$= 1 \cdot (\sqrt{3}+1) + 1 \cdot (\sqrt{3}-1)$$

$$= 2\sqrt{3}$$

(b) Find the angle θ between $\mathbf{v}$ and $\mathbf{w}$ by using the **Cosine** formula.

$$\text{Solution: } \cos \theta = \frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\| \cdot \|\mathbf{w}\|}$$

$$= \frac{2\sqrt{3}}{(\sqrt{2})(2\sqrt{2})}$$

$$= \frac{\sqrt{3}}{2}$$

$$\Rightarrow \theta = \frac{\pi}{6} \quad (\text{or } 30^\circ)$$

Problem 2. (10 points) Given $\mathbf{v} = (1, 1)$ and $\mathbf{w} = (1, 5)$. Choose a number c so that $\mathbf{w} + c\mathbf{v}$ is perpendicular to $\mathbf{v}$. (Hint: Use the relation $(\mathbf{w} + c\mathbf{v}) \cdot \mathbf{v} = 0$)

Solution: $(\mathbf{w} + c\mathbf{v}) \cdot \mathbf{v} = 0$

$$\mathbf{w} \cdot \mathbf{v} + c\mathbf{v} \cdot \mathbf{v} = 0$$

$$\Rightarrow c = \frac{-(\mathbf{w} \cdot \mathbf{v})}{\mathbf{v} \cdot \mathbf{v}}$$

$$\mathbf{w} \cdot \mathbf{v} = (1, 5) \cdot (1, 1) = 1 \cdot 1 + 5 \cdot 1 = 6$$

$$\mathbf{v} \cdot \mathbf{v} = (1, 1) \cdot (1, 1) = 1^2 + 1^2 = 2$$

$$\Rightarrow c = \frac{-6}{2}$$

$$= -3$$